# MAD 1 PROJECT REPORT - Placement Portal

## 1. Student Details

Name: Shashwat Anil Kumar Upadhyay
Course: BS Degree in Data Science and Applications
Roll Number: 23f3002413
Email: 23f3002413@ds.study.iitm.ac.in

## 2. Project Details

**Project Title: NexHire – Placement Portal Application**

### Problem Statement

Institutes often struggle to efficiently manage campus recruitment activities involving multiple companies and students. Traditional methods rely heavily on spreadsheets, emails, and manual coordination, making it difficult to track placement drives, manage applications, and avoid duplicate registrations.

To address this problem, the **NexHire Placement Portal** was developed as a centralized web application that allows **administrators, companies, and students** to interact on a single platform.

The system simplifies the placement workflow by enabling:

- Company registration and approval
- Creation and management of placement drives
- Student applications for drives
- Tracking of application status
- Administrative oversight of the entire recruitment process

This centralized system improves efficiency, transparency, and organization within the institute's placement management process.

## 3. Approach for NexHire Placement Portal

The system supports three primary user roles:

### Admin (Placement Cell)

The administrator is responsible for managing the overall placement system.

Admin functionalities include:

- Approving or rejecting company registrations
- Managing student and company record
- Approving placement drives created by companies
- Monitoring student applications
- Searching and managing users

### Company

Companies participate in recruitment by creating placement drives.

Company functionalities include:

- Company registration and profile management
- Creating placement drives with job details
- Viewing student applications
- Shortlisting or rejecting candidates
- Updating application status

### Student

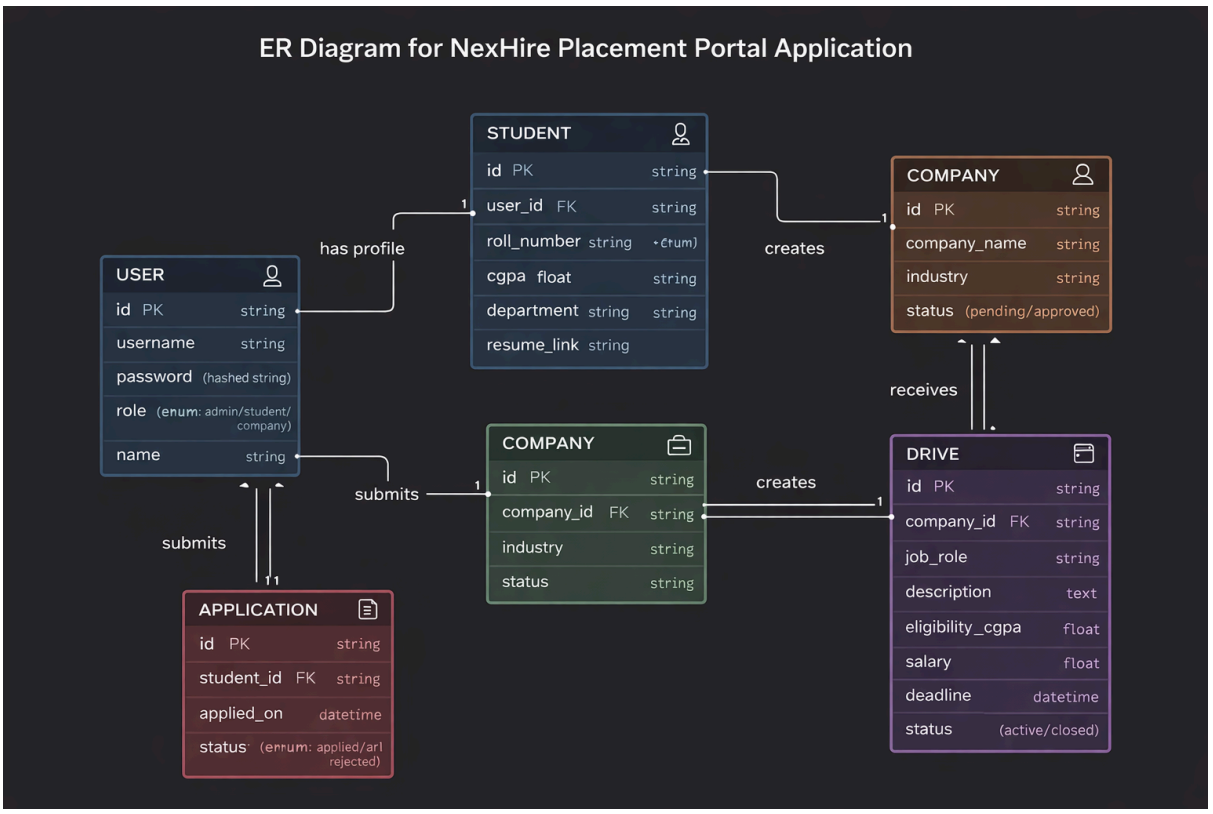
Students use the portal to apply for placement opportunities.

Student functionalities include:

- Registering and creating a student profile
- Viewing approved placement drives
- Applying for drives
- Viewing application status
- Tracking placement history

## 4. Database Architecture and Schema Design

The system's database utilizes a robust relational model designed to efficiently manage the placement workflow. As illustrated in the schema diagram, the architecture is divided into five core entities, intricately connected via primary and foreign key relationships to guarantee data integrity and seamless access control




- The **User** entity stores the authentication details of all system users. Each user has attributes such as username, password, role, and name. The role attribute determines whether the user is an administrator, student, or company representative.
- The **Student** entity stores student-specific information such as roll number, CGPA, department, and resume link. Each student profile is associated with a user account through a foreign key relationship.
- The **Company** entity stores details about registered companies including company name, industry type, and approval status. Only companies approved by the administrator are allowed to create placement drives.
- The **Drive** entity represents placement drives created by companies. Each drive contains job details, eligibility requirements, salary information, and application deadlines.
- The **Application** entity records student applications submitted for placement drives. It maintains information about the student who applied, the drive they applied to, the application timestamp, and the current status of the application.

# 5. Frameworks and Libraries Used

NexHire is built using the following modern web development technologies:

- **Flask:** The core backend framework handling routing, requests, and frontend-to-database communication.
- **Flask-SQLAlchemy:** An ORM library for interacting with the database using Python objects instead of raw SQL.
- **Flask-Login:** Manages secure user authentication, session handling, and role-based access control.
- **SQLite:** The lightweight database management system used to store all application data (users, drives, and applications).
- **HTML, CSS & Bootstrap:** Frontend technologies used to design a responsive, cross-device user interface.
- **Jinja2:** The templating engine that dynamically generates HTML pages using backend data.

# 6.AI / LLM Declaration

In the development of this project, several AI and design tools were used for assistance in coding, research, UI inspiration, and presentation preparation.

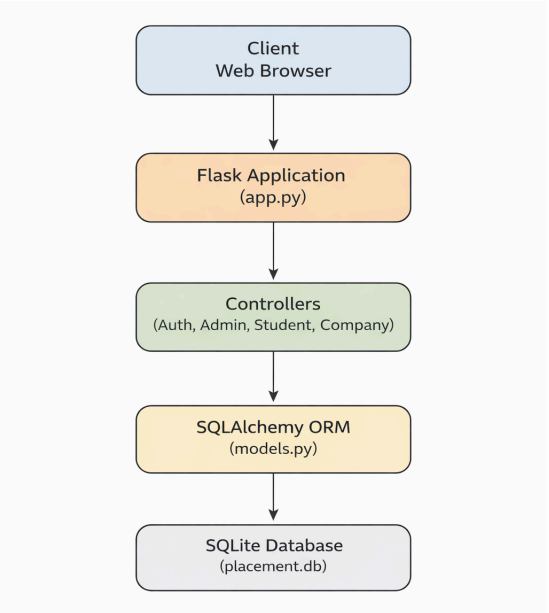
- **Gemini (Google):** Used for generating code snippets, structuring the Flask application using blueprints, debugging syntax errors, and assisting with documentation.
- **Perplexity AI:** Used for research and understanding best practices related to system architecture and placement portal workflows.
- **Dribbble:** Used as a reference platform for UI/UX design inspiration while designing the application's interface.
- **Canva:** Used for creating and editing the project presentation video and designing visual assets.

These tools were used only for assistance. The core logic, database design, implementation, and architectural decisions were developed and understood by the author to ensure the project fulfills the required specifications.

# 7. System Architecture

The NexHire system follows a layered architecture:

- **Client Layer:** Web browser used by users.
- **Application Layer:** Flask application handling requests and business logic.




- **Data Access Layer:** SQLAlchemy ORM for database interaction.
- **Database Layer:** SQLite database storing system data.

# 8.Implemented Features

1. **User Authentication System** – Secure login and registration for students and companies with role-based access control.
2. **Admin Dashboard** – Allows the administrator to manage students, companies, and placement drives while monitoring overall system activity.
3. **Company Registration and Approval** – Companies can register on the platform and must be approved by the admin before accessing system features.
4. **Placement Drive Management** – Approved companies can create and manage placement drives with job details, eligibility criteria, salary, and deadlines.
5. **Student Profile Management** – Students can register, update their academic details, and upload their resumes.
6. **Placement Drive Browsing** – Students can view all approved placement drives available on the platform.
7. **Application Submission and Tracking** – Students can apply to placement drives and track the status of their applications.
8. **Application Management for Companies** – Companies can view applicants and update the application status such as shortlisted, selected, or rejected.

# 9. Presentation Video

**Drive Link:**
https://drive.google.com/file/d/1eV6hyFYdih8nPERWxC4GMcw1Vvy9KSSp/view